

CYBERSECURITY

LAB 2

I. Introduction

The laboratory covers some of the basic cryptographic algorithms that were used for providing a secure way of communicating the messages between entities in the olden days. In these cryptographic algorithms, we assign numbers (or) algebraic elements to the given input message to be communicated between two entities. If the assigned numbers (or) algebraic values are in intelligent form, then it is considered as plaintext which is also called plaintext. This intelligible plaintext is converted into an unintelligible form called ciphertext. To convert the intelligible plaintext into the unintelligible ciphertext, an encryption function is used in the sender's side. Similarly, a decryption function is used in the receiver side to find intelligible plaintext from the unintelligible ciphertext. The process of converting the intelligible plaintext into unintelligible ciphertext and back into intelligible plaintext is called cryptography. This laboratory is discussing cryptanalysis of classical encryption algorithms such as *substitution*, *transposition*, and mixed algorithms.

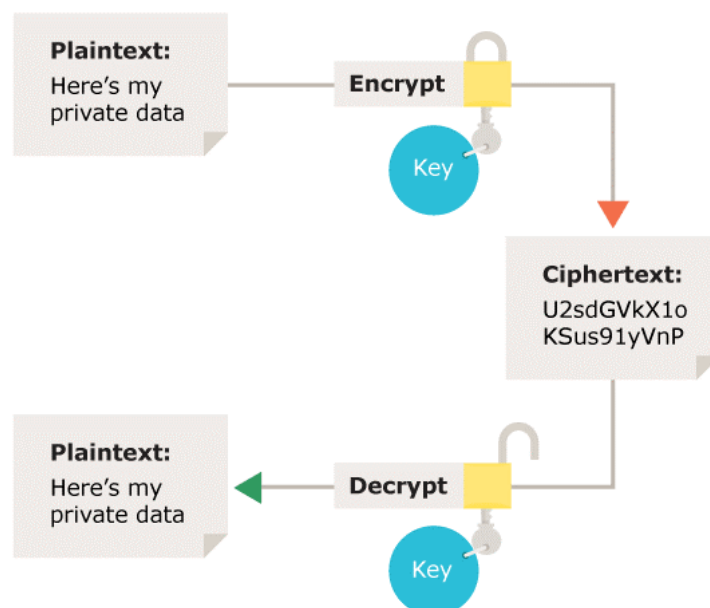


Figure 1 Encryption and decryption process. Source <https://ico.org.uk/>

Entropy is the amount of information in a given message or variable contains. Information entropy is sometimes described as “the number of bits required to communicate information.” From a mathematical point of view, it is the sum of the probabilities of a given sign and the number of occurrences within the data. It turns out that the level of entropy of different types of data is different. Thus, a text in natural language will have a different level of entropy than machine code, and this in turn will have a different level of entropy than compressed data. The entropy of an encrypted string of data, depending on the

encryption algorithm used, may not change at all or increase significantly.

Histogram shows relative frequency of each of the characters in the document. It allows us to compare the prevalence of different letters and can help derive the nature of the information and income cases even reveal some of the character's representation.

Autocorrelation is used to compare the text with its displaced copy and overlapping characters are counted. It turns out that if you move by the length of the key (or a multiple of it), there are many more overlapping characters. Also known as the coincidence index, it is an indicator of how likely it is that, when any two texts are compared letter by letter, the two characters being compared will be the same. Autocorrelation charts show the number of overlapping characters according to the offset of the text. Based on the autocorrelation graphs, the length of the used encryption key can be derived.

Periodicity searches for repeated fixed patterns in a document that start from any point and continue through to the end of the document. The pattern has two notable features, its offset and period. The offset is the start position of the cycle, counting from the first character of the document, and the period is the length of the set of characters that are repeated.

N-gram divides the message into blocks of N size and checks the frequency of phrases. It allows for evaluating the keys.

Floating Frequency breaks the message down into blocks of 64 characters and then displays the range of different letters for the block surrounding each letter. These results are then displayed to the user in the form of a line graph to allow them to see what parts of a message have high levels of variance and which have comparatively low variance. The purpose of this excursive is often to derive the nature of the data that has been encrypted. Frequency analysis is the basic tool for breaking most classical ciphers. In natural languages, certain letters of the alphabet appear more frequently than others. By examining those frequencies, you can derive some information about the key that was used. This method is very effective against classic ciphers such as Caesar, Vigenère, and so on. It is far less effective against modern methods, however. In fact, with modern methods, the most likely result is that you will get some basic information about the key, but you will not get the key.

II. Run and get familiar with the tool used in the laboratories - Cryptool.

- a. The CrypTool program is a free tool developed by the universities of Siegen and Darmstadt available at www.cryptool.org. The program is the basic platform used for several consecutive laboratory exercises. <https://www.cryptool.org/en/>
- b. The aim of the CrypTool creators was and still is to provide good quality teaching aids for illustrating and demonstrating the operation of selected mechanisms in cryptology.
- c. The program package includes several encryption algorithms, high-quality documentation on cryptography, and a demonstration of selected algorithms.
- d. Try to install and look at the capabilities of the program even before the classes.

- e. The first classes will mainly use the tools available in the Encrypt/Decrypt -> Symmetric (Classic) and Analysis -> Tool for analysis tabs.

Attention!

For exercises on historical algorithms, it is best to use input data in the form of a string of characters limited to the Latin alphabet, without special characters or punctuation.

For example:

DLACWICZENDOTYCZACYCHALGORYTMOWHISTORYCZNYCHNAJLEPIEJKORZYSTACZ
DANYCHWEJSCIOWYCHWPOSTACICIAGUZNAKOWOGRANICZONYCHDOALFABETULA
CINSKIEGOBEZ ZNAKOWSPECJALNYCHIINTERPUNKCYJNYCH

III. Tasks

1. Historical algorithms:

This section aims to learn about the operation of historical algorithms.

- 1.1. Check the available historical algorithms in tab *Encrypt/Decrypt* -> *Symmetric (Classic)*.
- 1.2. Choose a few of them and check their properties (space-number of keys, effect of multiple encryption, are there any weak keys e.g. for Rot-13 or Playfair)? Include the properties of the selected algorithms in the report.

Question:

- 1.3. What can we say about multiple encryption in the context of historical algorithms (consider their different classes)? How does this affect the ability to decrypt a ciphertext? The answer to this question is illustrated by the result of an experiment conducted in Cryptool.
- 1.4. Which of the selected algorithms do you consider the strongest, why?

2. Property analysis of available algorithms.

- 2.1. Compare the entropy values of plaintexts for different languages (English, Polish, German, French, Italian, Spanish, ...)
- 2.2. Compare the entropy values of plaintext and ciphertext depending on the algorithm.
- 2.3. In the task, one plaintext must be encrypted using the algorithms listed below:
 - Caesar,
 - adfgvx,
 - homophones,
 - permutations,

- vigenere (for different key lengths),
 - hilla (for different matrix sizes),
- 2.4. Compare histograms of plaintexts for selected 3 different languages (English, Polish, German, French, Italian, Spanish, ...) .
 - 2.5. Compare histograms of plaintext and ciphertext depending on the algorithm. For algorithms such as in point 2.
 - 2.6. Compare n-grams (bi/tri/n) of plaintexts for selected 3 different languages (English, Polish, German, French, Italian, Spanish, ...)
 - 2.7. Compare n-grams (bi/tri/n) of plaintext and ciphertext depending on the algorithm. For algorithms such as in point 2.
 - 2.8. Analyze the autocorrelation values of the ciphertext obtained by encryption XOR and Vigenere algorithms for different password lengths.

Questions:

- 2.9. How do the observed parameters change?
- 2.10. How can the text analysis tools available in CrypTool be used to determine the encryption algorithm for a given ciphertext?
- 2.11. How can the text analysis tools available in the CrypTool program be used to determine the password used for encryption?

3. Analysis of delivered files.

- 3.1. Try to recognize which encryption algorithm has been used based on an analysis of the ciphertext program only. All cryptograms were created based on the same plaintext. Files 1_X.txt,...
- 3.2. Try to decrypt by hand (using premises such as type algorithm, histogram, and other tools) or by using the automatic cipher-breaking option (Analysis - > Symmetric Encryption (Classic) -> Ciphertext-only) encryption programs placed in files 2_x.txt, ...
- 3.3. Recognize what encryption algorithm was used, and then try to decrypt the provided ciphertext programs. Files: 3_x.txt, ...

Questions:

- 3.4. What does the strength of the algorithm depend on?
- 3.5. How can the encryption strength be increased for known ciphers?